

# MOHAMMED SHABAZ S

Software Developer | Embedded Engineer

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## SUMMARY

Final-year Electronics & Communication Engineering student (CGPA 8.38) building across software and embedded systems. Internships at Nokia and iHelp Robotics, with projects in Android, computer vision, and microcontroller firmware using C, C++, Python and Embedded C. Currently deepening data structures, algorithms and SQL for software and embedded roles.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Java, SQL

**Software & Tools:** Android Studio, OpenCV, Firebase, Git, GitHub

**Embedded:** Embedded C, Microcontrollers

**CS Fundamentals:** Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks

## EXPERIENCE

### Student Intern

Sep 2026 – Present

Nokia Solutions and Networks India

Bangalore

- Working as a Student Intern at Nokia.

### AI Research Intern – Deep Learning & Model Development

Mar 2026 – Aug 2026

iHelp Robotics Private Limited · Project: MITRA

Remote

- Improved live camera-stream reliability in the MITRA assistive Android app across 2 video sources (hardware WiFi/RTSP and phone-camera fallback): cut stream search delay, persisted the last working transport, and added refresh/stall recovery.
- Engineered a visual-freeze watchdog plus 3 status/guard mechanisms — a stream-status panel, a cloud-status panel, and a WebSocket-gated send guard — and fixed a live-refresh loop that recurred after 5 minutes.
- Hardened offline voice commands (offline-preferred recognition with model recovery and microphone retry backoff) and verified builds across multiple devices with Gradle test/lint/assemble; delivered 6+ app-side improvements while model and backend stayed with a separate team.

### Student Intern – Skin Disease Classification

Jun 2026

IEEE EMBS Pune Section

Remote

- Trained and evaluated a skin-disease image classifier in a 3-member team during a 1-month IEEE EMBS internship — model selection, training, evaluation, and Python image-processing and testing; delivered a working demo-level workflow.

## PROJECTS

### VisionPay — Real-Time Indian Currency Detection

[GitHub](#)

Python, OpenCV, TensorFlow Lite, MobileNetV2, Text-to-Speech

- Developed an offline, real-time tool that recognizes Indian currency from a live webcam and speaks the result aloud for visually impaired users.
- Collected ~400 images per denomination across 6 classes (Rs.10–Rs.500) and trained a MobileNetV2 classifier converted to TensorFlow Lite, reporting ~93% validation accuracy.
- Improved reliability with 4 mechanisms: confidence/margin gating, temporal smoothing, a background class, and an auto-count mode.

### Automotive Body Control Module

[GitHub](#)

ESP32, Embedded C, GPIO, OLED, State Machine

- Programmed an ESP32 body-control module with a 3-state (OFF/ACC/ON) ignition FSM driving 6+ functions: indicators, synchronized hazard, brake logic, buzzer and an OLED dashboard.
- Architected a non-blocking millis()-based scheduler with brake debouncing, GPIO abstraction and edge-triggered logging for predictable real-time behavior.

### Smart Wellness Desk Assistant

[GitHub](#)

STM32 Nucleo-L476RG, STM32 HAL, Embedded C, Sensors · College team project

- Interfaced 2 sensor types (ultrasonic presence and ADC temperature), an I2C OLED, timers and buzzer alerts on an STM32 Nucleo-L476RG as part of a college team; handled wiring, testing and debugging in STM32CubeIDE.

### AEGIS — AI-Assisted Emergency Response Platform

Team lead · Architecture / design stage (in progress)

- Leading a 3-member team on an original concept, designing a 15-stage response architecture — from incident reporting and GPS/camera/audio inputs through AI analysis, human-dispatcher verification, resource dispatch and live tracking.

## EDUCATION

### B.E. Electronics & Communication Engineering

Expected 2027

Ballari Institute of Technology and Management (BITM), Ballari

Final Year · CGPA 8.38

Class 12: 80% · Class 10: 86.88%

## LEADERSHIP & ACTIVITIES

- Vice-Chair, IEEE CAS Society, IEEE Student Branch BITM (promoted from Treasurer); Treasurer, BITM Robotics Club.
- Google Student Ambassador (2025); Selected Participant, IEEE SPACE 2026 — B.Tech Initiative.

## CERTIFICATIONS

Embedded Systems — Internshala Trainings (2025) · Python Programming — EISystems (2024) · Google Cloud Generative AI — SmartBridge/SmartInternz (2025) · SQL for Data Analytics with AI · Programming in C and C++ with AI

## LANGUAGES

English · Kannada · Hindi · Urdu